B. Because our metabolic process is low.
C. Because much water is lost through sweat.
D. All of the above are correct

Explanation: In hot weather much water is lost from the body through the sweat. That is why we produce less amount of urine in such a condition.

Answer: C

107. Which of the following is the function of the liver?
A. Excretion of excess salt
B. Deamination of amino acids
C. Excretion of urea
D. Maintaining water balance

Explanation: Liver is an organ that has several functions in the body. Among the given alternatives liver is involved in deamination of amino acids.

Answer: B

108. Which of the following process do not take place in the liver?
A. Deamination
B. Detoxification of poisons
C. Storage of insulin and glycogen
D. Storage of insulin and glycogen

Explanation: Liver is involved in Deamination, Storage of glycogen, Detoxification of poison.

Answer: D

109. The two functions of the liver are:
A. Secretion of bile and storage of glycogen
B. Secretion of hepatin and storage of proteases.
C. Secretion of insulin and storage of hapatin
D. Secretion of insulin and glycogen

Explanation: Liver secretes bile and stores glycogen.

Answer: A

110. Which of the following excretory product is made in the liver?

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A. Minerals B. Urea C. Amino acids D. Glucose

Explanation: Urea is the excretory product made in the liver s a result of deamination of excess amino acids.

Answer: B

111. Which of the following is the principal organ for maintaining the body's water homeostasis?

A. Kidneys B. Skin C. Liver D. Lungs

Explanation: Skin, lung and kidneys play role in water balance. However, the principal organ where most of the osmoregulation(water balance) occurs is the kidney.

Answer: A

112. In which part of the nephron does ultrafiltration take place?

A. Bowman's capsule C. Distal convoluted tubule

B. Collecting ducts D. Loop of henel

Explanation: Ultra (pressure) filtration takes place in a part of a nephron known as Bowman's capsule.

Answer: A

Unit eleven review exercise

III. Multiple choice item

1. The correct path of nerve impulse in a reflex arc is:

A. Effector - sensory neuron → association neuron → receptor - motor

B. Effector → motor neuron - receptor → association neuron

C. Sensory neuron → receptor - Effector - motor neuron → association

D. Receptor → sensory neuron → association neuron - motor neuron

2. The correct order of ossicles from outer to inner ear is:

A. Stapes → malleus → Incus C. Malleus - Stapes → Incus

B. Malleus → Incus → Stapes D. Incus - malleus → Stapes

3. The part of the retina which contain light sensitive cells is:

A. Aqueous humor C. Vitreous humor

B. Yellow spot D. Blind spot

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4. In a depolirized neuron?
   A. The outside of axon membrane is positively change
   B. The inner side of axon membrane is positively change
   C. The inner side of axon membrane is negatively change
   D. There is no charge deference across the axon membrane.
5. The cells of the retina that are sensitive to bright light are:
   A. Sclera B. Cons C. Rod D. Chroid
6. The refraction of light in the eye ball is prevented by:
   A. Sclera C. Pupil
   B. Vitreous human D. Chorvid
7. Which of the following is true of a resting neuron?
   A. Its outside membrane becomes negative
   B. It has an action potential
   C. It is polarized
   D. It is depolarized
8. In human urinary system, the temporary storage sit of urine is:
   A. Bladder B. Urete: C. Urea D. Nephron
9. A hormone released by pituitary gland to regulate water balancing process in the kidney is:
   A. TSH B. FSH C. LH D. ADH
10. Which of the following is not part of the nephron?
   A. First convoluted tube C. Bowman's capsule
   B. Loop of Henle D. Renal tubule
11. Of the following substance which is present in the glomerular filtrate but normally completely reabsorbed befor the filtrate passes along the tube is:
   A. Sodium ion C. Urea
   B. Glucose D. Protein
12. Filtrate from glomerulus is first moves into:
   A. Loop Henle B. Collecting duct

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C. Proximal convoluted tubule D. Distal convoluted tubule

13. Which of the following structures of the urinary takes system connects the kidney with the bladder?

A. Urethra C. Renal vein

B. Loop of henel D. Ureter

14. In which part of the nephron does ultrafiltration takes place?

A. Bowman's capsule C. Distal convoluted tubule

B. Collecting duct D. Loop of henel

15. Which of the following is NOT found in the glomerular filtrate?

A. Plasma protein C. Glucose

B. Salt D. Water

16. All of the following are endocrine glands, EXCEPT the:

A. Thyroid gland C. Pituitary gland

B. Sweat gland D. Adrenal gland

17. The endocrine gland (s) that sits on top of the kidneys and secrets both metabolic stabilizers and stress regulators:

A. Thyroid gland C. Pituitary gland

B. Sweat gland D. Adrenal gland

18. The endocrine gland (s) referred to as the "master gland" is the:

A. Thyroid gland C. Pituitary gland

B. Pancreas D. Adrenal gland

19. The pituitary gland lies in the:

A. Abdominal cavity C. Inferior aspect of the brain

B. Along the femoral artery D. In the tissues of the neck

20. Name the gland that is located at the base of the throat.

A. Hypothalamus C. Pituitary

B. Pineal gland D. Thyroid

21. Which gland controls basal metabolic rate?

A. Parathyroid C. Pancreas

B. Thyroid D. Testes

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22. Which of the following hormones responsible for the "flight - or - flight" response?
A. Adrenaline
B. Glucagon
C. Insulin
D. Thyroxine

23. The pituitary hormone that stimulates are the male testes to produce sperm and stimulates the development of the follicle in the female on a monthly cycle is:
A. Growth hormone
B. Luteinizing hormone
C. Protraction
D. Follicel - stimulating hormone

24. Calcium level in the blood is regulated by the:
A. Thyroid
B. Pancreas
C. Parathyroid
D. Adrenal medulla

25. Which of the following is both an exocrine and endocrine gland?
A. Adrenal gland
B. Pituitary gland
C. Thyroid gland
D. Pancreas

26. The hormone that covert excess blood glucose level into glycogen is:
A. Insulin
B. Thyroid
C. Adrenaline
D. Glucagon

27. The target organ for insulin and glucagon is:
A. Kidney B. Brain C. Liver D. Testis

28. Which of the following hormones is secreted by pituitary gland?
A. Progesterone
B. Leutenizing hormone
C. Thyroxine
D. Parathyroid hormone

29. The hormone whose deficiency results in slow growth and stunted mental development in children is:
A. Adrenaline
B. Insulin
C. Thyroxine
D. Parathyroid

30. The endocrine gland which when surgically removed may result in disturbance of calcium level is:
A. Pituitary gland
B. Lselets of Langerhan's
C. Parathyroid gland
D. Adrenal gland

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31. What causes goiter?

A. Excessive protein intake
B. Iodine deficiency
C. Excessive salt intake
D. Vitamin deficiency

32. What is the cause of diabetes mellitus?

A. More production of adrenaline by adrenal gland.
B. Less production of insulin by pancreas
C. More production of progesterone by ovary
D. Lessor on production of bile by the liver.

33. Which of the following statement is correct about exocrine glands?

A. They are ductless
B. Secrete hormone directly into blood
C. Have duct
D. Secrete enzymes directly into the blood

34. Which of the following structures produces hormone

A. Organ B. Vessels C. Gland D. Neuron

35. Which of the following statement is true about exocrine gland?

A. It has special tube to carry its secretion
B. It is ductless and controls reproduction
C. Secretes hormone directly into the blood
D. Includes pituitary and thyroid gland

36. The deference between endocrine and exocrine gland is that endocrine gland are _____?

A. More in number in our body than exocrine gland
B. Have ducts whereas exocrine gland do not have ducts.
C. Do not have ducts; whereas, exocrine glands have ducts
D. Release their secretion through ducts to size of destination

37. The part of the ear that contain receptors for hearing is _____

A. Cochlea
B. Ossicles
C. Oval window
D. Semicircular canal

38. The correct match of endocrine glands and the hormones they produce

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is shown in:

A. Pancras - Insulin
B. Adrenal gland - glucagon
C. Thyroid gland - Parathyroid
D. Pituitary gland - Thyroxine

39. An impulse is carried from the inner ear to the brain by:

A. Optic nerve
B. Auditory nerve
C. Olfactory nerve
D. Effector neuron

40. What are the two bones of the middle ear that fit to the eardrum and the oval window respectively?

A. Stapes and anvil
B. Malleus and anvil
C. Satpes and incus
D. Malleus and stapes

41. Which of the following ear structures collects sound waves and directs in to the ear drum?

A. Middle ear B. Malleus C. Cochlea D. Pinna

42. Gray matter of the brain consists of:

A. Cell bodies and axon of the neuron
B. Dendrite of neurons and synapse lob
C. Cell bodies of neurons and synapse
D. Axon and dendrites of a neuron

43. The part of the brain that is concerned in coordination such as thinking, speech, memory, intelligence is:

A. Cerebellum
B. Cerebrums
C. Hypothalamus
D. Medulla oblongata

44. All of the following are the functions of hapothalamus, except:

A. Regulation of body temperature
B. Fat metabolism
C. Feeling of hunger and thirst
D. Balance and posture

45. The vision centre in the brain is:

A. Cerebrum
B. Medulla oblongata
C. Thalamus
D. Mid brain

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46. Voluntary movements of the body muscle are controlled by:

A. Cerebrum
B. Medulla oblongata
C. Optic lobe
D. Cerebellum

47. Vital centre for controlling heartbeat, respiration, blood, pressure and breathing is:

A. Medulls oblongata
B. Cerebrum
C. Hypothalamus
D. Cerebellum

48. Which of the following sense organs is incorrectly matched with the kind of stimulus detected by their receptors?

A. Eye - light
B. Tongus - Chemical
C. Nose - Mechanical
D. Ear - mechanical

49. All of the following are the protective structure of the eye except

A. Conjunctive
B. Eye socket
C. Eye lid
D. Vertebrae

50. The amount of light that enters the eye ball is regulated by

A. Cornea B. Lens C. Pupil D. Retina

51. In dim light :

A. Pupil gets constricted
B. Circular muscles contract
C. Radial muscle contract
D. Radial muscles relax

52. The ability of the eye to see near and distant objects is called:

A. Focusing
B. Seeing
C. Accommodation
D. Observation

53. Which of the following ear bones directly transmits sound to the cochlea?

A. Incus B. Ampullae C. Stapes D. Malleus

54. Receptors that detect body position are found in the:

A. Eye B. Ear C. Skin D. Nose

55. Endocrine system differs from a nervous system because:

A. It uses electrochemical impulse transmission
B. Uses chemical coordination systems

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C. Messages are transmitted through neurons
D. Provides quick response.

56. A type of neuron that carries an impulse to the central nervous system is:
A. Efferents neuron
B. Relay neuron
C. Motor neuron
D. Afferent neuron

57. Structures found on sense organs and can detect stimulus are:
A. Efferents B. Gland C. Receptors D. Neurons

58. Neuron at rest is called:
A. Repolarized
B. Depolarized
C. Deactivated
D. Polarized

59. _____ is the wave of depolarization across the neuron
A. Resting potential
B. Axon potential
C. Action potentials
D. Nerve potential

60. A junction between two neuron is:
A. Dendrite
B. Axon
C. Mylean sheath
D. Synapse

61. All of the following are the protective structures of the brain, except:
A. Cranium
B. Vertebral column
C. Meninges
D. Cerebrospinal fluid

62. Nerves that are directly connected to the brain are:
A. Cranial nerves
B. Sensory nerves
C. Spinal nerves
D. Motor nerves

63. Which of the following is the structural and functional unit of the human nervous system?
A. Brain B. Axon C. Dendron D. Neuron

64. What does corpus callosum connects in human brain?
A. Two optic lobes
B. Two cerebral hemisphere
C. Bone and muscle
D. Two lobes of pituitary gland

65. Which of the following part of the human brain has a center for

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controlling breathing?

A. Medulla oblongata
B. Diencephalon
C. Cerebrum
D. Cerebellum

66. How many pairs of the spinal nerve are found in human?

A. 33 B. 31 C. 13 D. 12

67. Which of the following is a balancing organ in humans?

A. Eardrum
B. Semi - circular canal
C. Cochlea
D. Ear canal

68. Which of the following cells in the pancreas make insulin?

A. Epithelial cells
B. Hepatic cells
C. Germ cells
D. Islet cell

69. Which of the following is the largest gland in human body?

A. Thymus B. Kidneys C. Liver D. Pancreas

70. Which of the following human cells produce insulin?

A. Beta cells B. F cells C. Alpha cells D. Delta cells

71. Which of the following is the inner most layer of human eye?

A. Retina B. Choroid C. Sclera D. Cornea

72. Which of the following control breathing rate in humans?

A. Cerebellum
B. Hypothyalamus
C. Thalamus
D. Medulla oblongata

73. Which of the following disease not related to thyroid gland?

A. Myxoedema
B. Cretinism
C. Acromegaly
D. Goiter

74. Which of the following is not the symptom of hypothyroidism?

A. Loss of bone mass
B. Cretinism
C. Myxedma
D. Goiter

75. Endemic goiter is a state of _____

A. Increased thyroid function
B. Decreased thyroid function
C. Moderate thyroid function
D. Normal thyroid function

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76. The four small glands in the thyroid are called _____
A. Parathyroid gland
B. Adrenal gland
C. Pineal gland
D. Endocrine and exocrine gland

77. FSH and LH are collectively known as _____
A. Neurohormones
B. Antistress hormones
C. Gonadotrohic hormone
D. Emergency hormone

78. Which of the following hormones is a steroid?
A. Epinephrine
B. Thyroxine
C. Estrogen
D. Prostaglandins

79. Secretion of many anterior pituitary hormones are controlled by other hormones from _____
A. Pancreatic lobes
B. Adrenal gland
C. Hypothalamus
D. Thyroid gland

80. _____ is a quick response to the stimuli that passes the brain.
A. Reflex action
B. Voluntary action
C. Knee jerk
D. Receptors

81. The cerebellum is located between the cerebrum and brain stem in the back of the head. It helps in _____
A. Breathing and controlling blood pressure
B. Balance and coordination
C. Voluntary movement
D. Speech and hearing

82. Which connects two hemispheres of the brain?
A. Pons
B. Pia matter
C. Corpus callosum
D. Diencephalon

83. Fluid filled cavity on the brain contains _____
A. Matter
B. Cavity
C. Meninges
D. Diencephalon

84. Which part of the brain controls higher mental activities like reasoning?
A. Temporal lobe
B. Frontal lobe

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C. Medulla oblongata D. Cerebellum

85. In CNS myelinated fibres form the _____ while non myelinated fibers cells from the _____

A. Grey matter, white matter

B. Ependymal cells, neurosecretory cells

C. White matter, grey matter

D. Neurosecretory cells, Ependymal cells

86. How many pairs of cranial nerves originate from the brain?

A. 8 B. 31 C. 1 D. 12

87. Skeletal muscles are controlled by _____

A. Somatic nerves C. Parasympathetic nerves

B. Autonomic nerves D. Sympathetic nerves

88. The thin and convoluted outer layer of gray matter that covers the cerebral hemisphere is _____

A. Meninges C. Medulla oblongata

B. Thalamus D. Cerebral cortex

89. In peripheral nervous system the nerves that arise form the brain are celled as _____

A. Spinal nerves C. Temporal nerves

B. Cranial nerves D. Frontal nerves

90. Spinal cord and brain are wrapped in protective membranes known as _____

A. Nodes of Ranvier C. Axon membranes

B. Meninges D. Mylein sheath

91. Myelinated neuron has

A. Cells bodies

B. Dendrites

C. A rapid communication system

D. A slow communication system

92. Nervous system is _____

A. An external communication system

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B. A slow communication system
C. A rapid communication system
D. A moderate communication system

93. Organ of cochlea helps in _____
A. Balancing
B. Maintaining equilibrium
C. Hearing
D. Formation of wax

94. A person goes to ophthalmologist. He has a problem of reading books because he is not able to contract his _____
A. Iris
B. Suspensory ligament
C. Pupil
D. Ciliary muscles

95. Which of the following prevents internal reflection of light inside the eye?
A. Cornea B. Sclera C. Choroid D. Retina

96. Sense of smell is perceived by _____
A. Pituitary
B. Hypothalamus
C. Olfactory lobe
D. Cerebrum

97. Conversion of vibrations into compressed wave takes place in which part of human ear?
A. Inner ear B. Middle ear C. Outer ear D. Exterior ear

98. Which of the following is known as the "window of brain"?
A. Sensory organs
B. Cranial nerves
C. Eyes
D. Ganglia

99. Which is the common problem with the endocrine system?
A. Bad gas
B. Diabetes
C. Hemophilia
D. Lung cancer

100. What is the function of ADH?
A. Controls sugar level in blood
B. Decreases water absorption
C. Increases water absorption
D. Synthesis of salt

101. The pressure in the middle ear is equalized by:
A. Ear canal
B. Ossicles

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C. Pinna D. Eustachian tube

102. A widely used and frequently abused psychoactive drug is?
A. Alcohol B. Cigarette C. Khat D. Cannabis

103. The name of drug in khat is _____ ?
A. Ethanol B. Nicotine C. Cathinone D. Arsenic

104. The type of drug grouped under depressant is _____ ?
A. Cocaine B. Alcohol C. Cannabis D. Coffee

105. Khat is absorbed into blood stream through the membrane of ____ ?
A. Mouth and the stomach C. Nose and mouth
B. Intestine and skin D. Stomach and intestine

IV. Short answer

106. The process of maintaining constant internal condition of the body is called _____

107. Animals with variable body temperature are celled _____

108. The process of regulating the water balance and ionic control of the body is called _____

109. _____ are clusters of capillaries in the Bowma's capsulle.

110. The process removing excess amino acids from the body is called _

111. A fatty insulating layer around the around the axon is called _____

112. _____ is a wave of depolarization across the neuron.

113. The layer of meninges that adhere tightly to the surface of the brain is _____

114. The type of cells that support and nourish the neuron is known ____

115. _____ Is a person uses a substance to the point of excess and/ or dependence

116. The central nervous system consists of _____ and _____

117. _____ is a neural pathway during reflex action.

118. A sudden, automatic and uncontrolled response of the stimuli is called ____

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V. Essay type:

119. List the major parts of the neuron?
120. Explain the difference between a neuron and a nerve?
121. What is a role of neurotransmitters in nerve impulse transmission?
122. Explain the meaning of action potential.
123. List some function of cerebrospinal fluid (CSF)?
124. List the three types of cells found in the epidermis of a skin and explain their function?
125. ____ is a tiny projection on the tongue, which is rough texture and on which the taste buds are located?
126. Explain the importance lacrimal gland of the eye?
127. What is he main difference between taste and smell receptor?

Answer to unit review exercise

I. Multiple choice

1. D	11.D	21.B	31.B	41.D	51.C	61.B	71.A	81.B	91.C
2. B	12.C	22.A	32.B	42.A	52.C	62.A	72.D	82.C	92.C
3. B	13.D	23.D	33.C	43.B	53.C	63.D	73.C	83.D	93.C
4. B	14.A	24.C	34.C	44.D	54.B	64.B	74.A	84.B	94.D
5. B	15.A	25.D	35.A	45.A	55.B	65.A	75.B	85.C	95.C
6. D	16.B	26.A	36.C	46.D	56.D	66.B	76.A	86.D	96.C
7. C	17.D	27.C	37.A	47.A	57.C	67.B	77.C	87.A	97.A
8. A	18.C	28.B	38.A	48.C	58.D	68.D	78.C	88.A	98.A
9. D	19.C	29.C	39.B	49.D	59.C	69.C	79.C	89.B	99.B
10.D	20.D	30.C	40. D	50.C	60.D	70.A	80.A	90.B	
100.C		102.A		104.B					
101.D		103.C		105.A					

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II. Short Answer

106. Homeostasis
107. Poiklotherms
108. Osmoregulation
109. Glomerules
110. Deamination
111. Myelien sheath
112. Action potential
113. Pia matter
114. Glial cell
115. Drug (or substance abuse)
116. Brain & spinal cord
117. Reflex are
118. Reflex action

III. Essay type:

119. A neuron has the following major parts: \Axon, Cell body, Dendrite
120. A neuron is a single nerve cell while a nerve is a bundle of neurons.
121. Neuro transmitters are chemicals that carry an impulse along the axon
122. Action potential is a wave of depolarization across the neuron
123. Absorb shock, support weight of the brain, protect the brain from damage, and maintain uniform pressure around the brain, exchange nutrient and waste
124.
    a) **Keratinoctytes** - produce protein (keratin) to make hair, nail surface of skin
    b) **Melanocytes** - produce pigment called melanin
    c) **Langerhans cells** - prevent water loss and entry of pathogens.
125. Papillae.
126. Produces tear, containing lysozyme, which kills bacteria. The tear washes away dust particles.
127. The major functional deference between taste and small receptors are:
    - **The smell receptors** are more specialized for detecting vapors coming to the organism from distant sources but taste receptors are specialized for detection of chemicals present in the mouth itself.
    - **Smell receptors** are much sensitive than taste receptors.

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UNIT 12

12. Climate Change

What is climate change ?

Climate change refers to long - term shifts in temperature and weather patterns. This shift is mostly due to human activates, such as burning of fuel, which produces heat - trapping gases.

It is the global phenomenon of climate transformation characterized by the changes in usual climate of the planet that are particularly caused by human activities.

It is systematic change in the long - term system of the atmosphere over multiple decades or longer

- The surface of the earth has warmed by 0.7 ^{0}C over the century.
- There are fears that this warming is increasing

12.1 What are the Causes of Climate Change?

Human Factor:

The human activities, such as industrialization, transportation urbanization, over population, through:

- Burning fossil fuels
- Cutting down forests
- Farming livestock that increase the concentration of atmospheric CO_2 and methane gases

Natural factors:

- Solar radiation
- The change in the amount of solar energy that enters the earth and reflected out space. Much of the energy is trapped and heats atmosphere.

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➥ Volcanic activities, wild fire releases CO_2 into the atmosphere

The possible consequences of climate change could be a rise in sea in level, flooding, drought, reduction in crop production

12.2 What are Greenhouse Gases (GHGs)?

GHGs are gases in the earth's atmosphere that trap heat.

The gases act like glass walls of a greenhouse and hence are called **greenhouse gases.**

The major greenhouse gases are:

- ➥ Water vapor
- ➥ Carbon dioxide (CO2)
- ➥ Methane
- ➥ Nitrous oxide.
- ➥ CFCs

CO_2 is the greenhouse gas responsible for the greatest amount warming. Because:

- ➥ It absorbs less heat per molecule than methane or nitrous oxide, but it is more abundant and stays in the atmosphere much lomger.
- ➥ It is transparent to sunlight, but traps heat
- ➥ Responsible for about 2/3 of total energy imbalance that is causing the rise of temperature on earth

The abundances of greenhouse gases in the atmosphere are fewer than that of oxygen and nitrogen, but GHGs absorb the heat and release it gradually overtime to rise earth's temperature.

The natural balance of CO_2 concentration is maintained through:

- ➥ Photosynthesis
- ➥ Respiration
- ➥ Decomposition.
- ➥ An exchange of gases between ocean and atmosphere

The imbalance of CO_2 in the atmosphere is due to an increase in fossil fuel combustion and deforestation.

In 21 century the level of CO_2 is 40% higher it was in 19th century.

- ➥ This is mostly due to combustion of fossil fuel.
- ➥ Deforest has also contributed

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12.2.1 What are the Effects of Greenhouse Gases ?

Solar radiation passes through the atmosphere and warms the earth's surface

- The earth radiates heat energy back into space, some of the energy is absorbed by the greenhouse gases.
- It acts as a blanket around earth and prevent some heat energy escaping into space
- This cause the air warm up

Because of greenhouse gases, temperature of the earth is about an average of $14C^{o}$.

With the effect of greenhouse gases, temperature of the earth would drop to $-18\ ^{o}C$ that can't support life.

GHGs are sometimes "called climate active pollutants" because of the effects on human health.

The following table show greenhouse gases and their sources.

Types of greenhouse gases:	The sources greenhouse gases
Carbon dioxide (CO_2)	Fossil fuel combustion, deforestation, volcanic eruption, respiration and wild fire
Methane (CH4)	Landfills waste damp, rice farming, gaming, digestive tracts of cattle, sheep, decomposition, bio gas production
Nitrous oxide (N_2O)	Fertilizers, animal waste, biomass burning, fossil fuel combustion
Hydroflorocarbons (HFCs)	Refrigerator, coolant, semiconductor manufacture and industrial process
Perflurocarbons (PFCs)	Refrigerator, coolant, semiconductor manufacture and industrial process and aluminium smelting
Sulfur hexafluoride (SF6)	High voltage electrical transmission system (23, 000x global warming potential than CO_2), magnesium and aluminium production

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HFC, PFC and SF6 are totally man - made during industrial process and don't occur in nature.

Water vapor is the most abundant greenhouse gas.

- It increases as the earth's atmosphere warms.
- But unlike CO2, it persist in atmosphere for only a few days.

12.3 What are the Effects of Climate Change?

Climate change destabilize the earth's temperature balance, affects human being and the environment. It can be direct or indirect effects:

I. Direct consequence of man - made climate change

- Rising maximum temperatures
- Rising sea levels
- Higher ocean temperature
- An increase in heavy precipitation and shrinking of glaciers.
- Thawing of permafrost

II. Indirect consequences of climate change:

- An increase in hunger and water crises particularly in developing countries.
- Health risks due to the rising air temperature and heat weaves.
- Economic crisis.
- Increasing spread of pests and pathogens.
- Loss of biodiversity due to limed adaptability of flora and fauna.
- Ocean acidification due to increased HCO_3 as a result of high CO_2 in the water

Note: generally the effects of climate change can be seen in biodiversity, agriculture, forest productivity, natural disasters, health, and water resource.

12.3.1 The Effects of Climate Change in Biodiversity

Biodiversity is the variety of living things in an area, which provides immense direct benefits to humans.

Climate change is the direct driver of biodiversity loss: projected change in climate combined with land use cause:

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- The spread of exotic or alien species, which out compete the native species
- Some species migrate to other places and therefore will accelerate their loss.
- It affects habitats of many species so that they are forced to migrate to other places or adapt

40% of the world's economy being derived from biological resources.

Maintaining biodiversity provides:

- Greater food security
- Opportunities for economic development
- Foundation for new pharmaceuticals and other medical advances.

12.3.2 The Effects of Climate Change on Agriculture

Climate change can disrupt food availability, reduce access to food and affect food quality. Agricultural biodiversity refers to all components of biological diversity relevance to food agriculture. It includes:

I. Plants genetic resource such as:

- Crops
- Wild plants harvested and managed for food
- Trees farms
- Pastures
- Range land species
- Medicinal plants
- Ornamental plants of aesthetic value.

II. Animal genetic resources:

- Domesticated animals
- Wild animals hunted for food
- Wild and farmed fish and other aquatic organisms
- Insect pollinators

III. Microbial and fungal genetic resource:

It includes bacteria fungi

Generally, climate change can affect agriculture in terms of:

- Declining soil fertility,

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- Decreasing crop productivity
- Increasing water requirement by plants.
- Increasing desertification
- Decrease revenue, etc.

12.3.3 Effects of climate change on forest productivity

Climate change could alter the frequency and intensity of forest disturbances, hence reduces forest productivity and change the distribution of species. This is due to:

- Pest outbreaks, which defoliate the leaf, weaken and kill trees.
- Invasive species out compete the native tree species for water, food and space.
- Wildfire that occur during warm summer and drought season damage trees and forest.
- Storms, such as floods can damage the forest,

Some species of trees in the forest may recover, but others may change their range or die.

- However, once the structure of the forest is changed, it will never be like the existing one. Because:

The new species that colonize the area create a new type of forest.

12.3.4 Climate change and Natural Disasters

Climate hazards are natural events or agents of disaster in terms of what they may do to human settlements or to the environment in weather cycles. It includes:

- → Hurricanes
- → Droughts
- → Wildfires
- → High winds
- → fog
- → Flooding
- → Thunderstorms
- → Temperature extremes, etc.

Identification of hazardous events is not always easy. But it has certain criteria, such as:

- Property damage
- Economic loss
- Disruption of social service (eg. Communication failure)

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- Excessive strain in essential service such as police, fire

When more water vapor is accumulated into the atmosphere, it becomes ideal for more powerful storm to develop.

The effects of climate change and natural disasters:

I. Melting ice and rising seas

- Global warming causes polar ice sheets and glaciers to melt.
- Melting ice cause sea levels to rise resulting flooding erosion of coastal and lower lying areas.

II. Extreme weather and shifting rainfall

- Heavy rain and other extreme wealth events are becoming more frequent
- It leads to floods, and decreasing water quality and quantity in some regions.

III. Risks on human health

- Climate change is already having an impact on health.
- Climate change causes an increase in the number of heat - related deaths in some regions of the earth and a decrease in cold - related deaths in other parts of the world.
- Climate changes increases the spread of some water - borne illnesses and disease vectors.

Illustrative Example

1. Which of the following is not the definition of climate change?
Climate change:

A) Refers to long - term shifts in temperature weather patterns.

B) It is the global phenomenon of climate transformation

C) It is a systematic change in the long - term state of the atmosphere over multiple decades

D) Refers to a daily or monthly shift in temperature and weather patter

Explanation: a short term (daily or monthly) shift in weather such as temperature, wind or rainfall is called weather condition. Climate change is long - term shifts in temperature and weather patterns.

Answer: D

2. What is the term for the natural ability of Earth's atmosphere to trap energy from the sun?

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A) Global warming
B) Greenhouse effect
C) Ozone depletion
D) Biological depletion

Explanation: the greenhouse effect is the ability of some gases such as CO_2, methane, etc. to trap radiation energy of the sun

Answer: B

3. Carbon dioxide is called a greenhouse gas because:

A) It is used in greenhouses to increases plants growth
B) It is transparent to sunlight, but traps heat.
C) It is transparent to heat, but traps sunlight.
D) It is transparent to both sunlight and heat.

Explanation: CO_2 is a transparent molecule, but traps radiation heat of sun in the atmosphere.

Answer: B

4. How much is CO_2 accounts for the energy imbalance in the atmosphere and rise of temperature among other gases?

A) 70% B) 50% C) 25% D) 100%

Explanation: CO_2 is responsible for about 2/3 of total energy imbalance that is causing the rise of temperature on earth.

Answer: A

5. Climate change occurs because:

A) Carbon dioxide and other greenhouse react chemically to produce excess heat
B) Earth has too many greenhouse and other glassed buildings
C) Volcanic eruptions produce large quantities of sulfur and other greenhouse gases
D) Carbon dioxide and other greenhouse gases trap infrared radiation in the gases

Explanation: Climate change is occurred mainly due human activities by releasing greenhouse gases such as CO_2, CH_4, nitrous oxide etc. into the atmosphere which traps inferred radiation of the sum.

Answer: D

6. Which of the following is the most abundant gas in the atmosphere?

A) Oxygen
B) nitrous oxide

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C) Carbon dioxide D) Methane

Explanation: The most abundant gases in the atmosphere are oxygen and atmosphere

Answer: A

7. One of the following is not the natural way of balancing CO_2 in the atmosphere?

A) Respiration C) Decomposition

B) Photosynthesis D) Burning of coal

Explanation: The natural balance of CO_2 concentration in the atmosphere is maintained through:

- Photosynthesis
- Respiration
- Decomposition
- Ans exchange of gases between ocean and atmosphere.

Answer: D

8. List some examples of natural disaster, which occurs as a result of climate change?

Explanation: Melting ice and rising seas, extreme weather and shifting rainfall, and risks on human health.

9. Why CO_2 is responsible for the greatest amount global warming than other greenhouse gases?

Explanation: because CO_2 is least abundant but stays in the atmosphere much longer than the others.

10. Describe how climate change may affect sea level, precipitation patterns, organisms (including humans), and food production?

Explanation

- During the 21st century, climate change is causing a rise in sea level.
- Precipitation patterns are changing, resulting in more frequent droughts in some and more frequent flooding in other areas.
- Biologists think that climate change will cause some species to go extinct, some to be unaffected, ad others to expand their numbers and ranges.
- Problems for agriculture include increased flooding, increased

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droughts, and declining agriculture productivity in tropical and subtropical areas.

11. Why the heat trapping gases are termed greenhouse gases?

Explanation: because theses gases are act like glass walls of a greenhouse, which trap heat and hence are called greenhouse gases.

12. Discus the effect of the absence of greenhouse gas on life?

Explanation: without greenhouse gases, temperature of the earth would droup to - 18°C, that is too cold to support life.

12.4 Safety Rules and Precautions During Natural Disaster

Natural disasters are catastrophic events that often occur without warning, which results in:

- Disrupt the ecosystem
- Causing damage to person lives, property, transportation, and livelihood

During natural **disaster risk management** is a comprehensive approach involved the identification of threats through implementation of the proposed mitigation measures.

Natural disaster can't be prevented: but the damage can be minimized through timely preparation

Every natural disaster has its own set of precaution to be taken: priority must be given to save precious lives.

12.4.1 Natural Disaster and its Precaution

To keep you and your family safe during a natural disaster, the following general preparedness safety tips can prevent injuries and make the difference in an emergency:

- Stay informed
- Have a plan for evacuation
- Keep emergency kits on hand.
- Avoid unnecessary rist
- Go to the safest area in your home

Earthquake is the shifting of tectonic plates under the earth's crust, which are responsible for mass destruction. precautions during earth quake:

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- If you are indoors: take cover under a sturdy table or furniture until the shaking stops, stay away from glass, windows, door, wall, and anything that could fall (lighting or furniture). Stay inside until shaking stops: don't move inside the building or leave. Don't use elevator.
- If you are outdoors: stay away from building, steel light and utility wires: stay in open group. Most earthquake related causalities result from collapsing walls, flying glass, and falling object.

Tsunamis are a series of enormous ocean waves caused by earthquakes underwater, landslides or volcanic eruption. Precautions. during tsunami:

- Follow precautions instructions from radio or TV
- Move from inland to higher ground and stay there:
- Stay away from flooded and damaged area,
- Keep away from debris in the water because it may pose a safety hazard to boats and people.

Cyclones are tropical storm caused by atmospheric disturbances around a low - pressure area, it is accompanied by strong winds moving as a speed of sixty two kmph or more:

- Be alert to the change weather
- Listen to radio/TV for instruction
- Look for approaching storms
- Look for the following warning sign: dark, greenish sky, rotating cloud, roars.
- If you see approaching storm, take shelter immediately

Floods are among the earth's most common and dangerous natural hazards, which are formed due to a flow of water on areas of land that are usually dry. Excessive rains can damage nearby dams by forming flooding.

- Don't attempt to walk, swim, or drive through the floods.
- Floods may contain contaminations, thus take care of contamination
- Floods could be deadly due to fallen electrical line in the water.
- Stay clear of bridge over fast moving water.
- Keep an eye out for evacuation alert.
- Move to higher ground.
- If your car is trapped in flood and water start filling inside the car, seek refuge on the roof.

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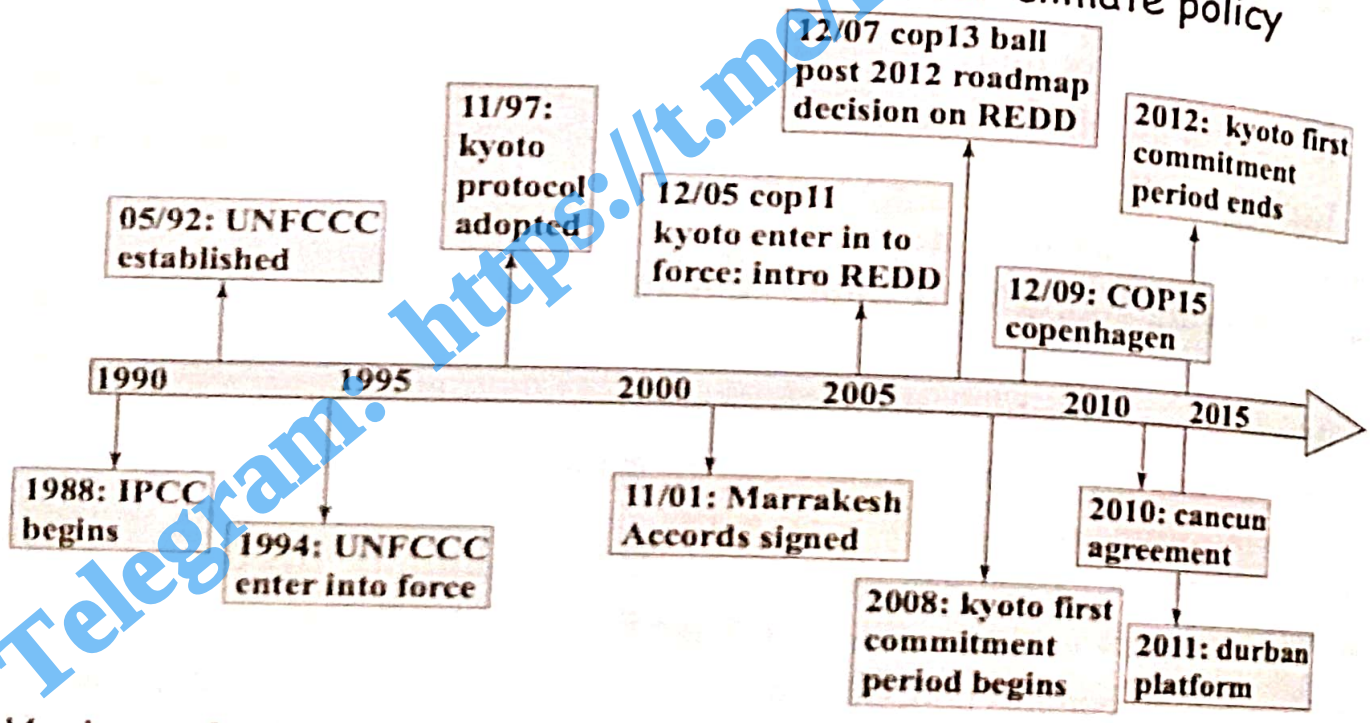
12.4.2 Disaster Mitigation

- Mitigation is defined as sustained action that reduces or eliminates long term risk to people and property from natural hazards and their effects.
- It is ongoing effort to lessen the impact of people and property
- They are means to prevent, reduce and control adverse environment effects.
- It involves keeping homes away from flood plain, engineering road and brigs to withstand earthquake, buying flood insurance, developing effective building standards.

12.5 International conventions on climate change

Climate change is a long - term global problem, which requires stable but flexible policy implementation over time.

The following chart shows the time line for climate policy



Various international conventions have been evolved. For example:

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12.5.1 The United Nations Framework Convention on Climate Change (UNFCCC)

The UNFCCC is an international environmental treaty agreed in 1992.

Its objective is to prevent dangerous man - made interference in the global climate system

For example, stabilizing greenhouse gas concentrations in the atmosphere

- Ethiopia is among the 197 Party member state of the Conversion

12.5.2 Kyoto Protocol on Climate Change

- Kyoto protocol was adopted on 11 December 1997, in Kyoto, Japan.
- It is the first international treaty to set legally binding target to greenhouse gases emission
- The Kyoto protocol operationalizes the UNFCCC by committing industrialized countries and economies in transition to limit and reduce greenhouse gas emission in accordance with agreed individual targets.
- The Kyoto protocol is an international agreement that called for industrialized nation to reduce their greenhouse emissions significantly.

The first agreement was in 1997 and takes 8 years for member state for ratification in 2005. The Kyoto protocol enters into force in 2005. For example

- Developed countries are legally obliged to reduce greenhouse gases emission by 5%
- Developing countries such as china, India, Brazil and South Africa have no restriction on gas emission, but encouraged to adapt policies to promote growth.

Kyoto offered market mechanism to help rich countries meet their target:

- They are encouraged to invest in low carbon emission project in poor countries.
- Through industrialized countries that emit less they are allowed to can sell this amount to industrialized countries that produce more than they should.

The ratification was hailed as "environmentally strong and economically sound" by US president Bill Clinton.

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12.5.3 Copenhagen (COP 15) in 2009

For the first time it was decided that each country would propose a national contribution know as Intended Nationally Determined Contribution (INDC) to reduce emission.

All the 195 UNFCCC countries promised to reduce their greenhouse gas emissions by 2025 - 2030.

It aimed to strengthen the global response to the threat of climate change, in the context of sustainable development and efforts to eradicate poverty.

It contains three key provisions.

1. Holding the global increase of average temperature bellow 2 °C up to 1.5 0C above pre - industrial level, which is believed to reduce the risk of climate change
2. Develop mechanisms or adaptation to the adverse effects of climate change.
   - ➥ Climate resilience development.
   - ➥ Low greenhouse emissions development in a manner that doesn't threaten food production.
3. Making finance flows consistent with a pathway toward low greenhouse emission and climate resilient development

12.5.6 The Paris Agreement

Signed in 2015, and interred into force in 2016, which is effectively replaced the Kyoto protocol.

Has 194 parties, requires all countries to reduce emissions.

- ➥ Including developing countries such as Chain, India, which were not included in Kyoto protocol

Government set targets, known as nationally determined contributions

- ➥ With the goals of preventing the global average temperature from rising more than 2°C above pre industrial level and pursuing efforts to keep temperature rise below 1.5 °C

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12.6 International and Natural Practices of Implementation of Conventions

Ethiopia is one of the world's most drought - prone countries, vulnerable to hydro - meteorological hazards, and other natural disasters.

- The climate changes have been affecting rain - fed agriculture, water, forestry, and poverty level.
- Ethiopia's policy response to climate change has progressively evolved since the approval of the UNFCC in 1994.
- Ethiopia launched the National Adaptation plan of Action in 2007 and the Ethiopia program of Adaptation on Climate Change and Nationally appropriate mitigation Actions in 2010.
- More over Ethiopia endorsed a Climate Resilient Green Economy (CRGE) strategy in 2011 to build a green and resilient economy.
- Over the years, Ethiopia has been implementing various programs within those policy frameworks. For example:

i. The Ethiopia green legacy initiative:

- The green legacy imitative was lunched in 2019 with the vision of building a green and climate resilient Ethiopia.
- Targeted to plant 20 billion seedlings within a period of four years.
- The Green Legacy Initiative is a demonstration of Ethiopia long - term commitment and responsive and climate change and environmental degradation.
- The Green Legacy Initiatives encompasses agroforestry, green and renewal of urban areas and integrated water and soil resource management.

ii. The Grand Ethiopia Renaissance Dam (GERD):

- It is the best examples in Ethiopia to support climate resilience green economy
- It is a hydroelectric dam on the Abay (Blue Nile) river.
- The dam is located in Benishangul Gumuz region
- It is the largest hydroelectric power in Africa
- It is under construction since 2011
- The primary purpose is to generate electricity with a capacity of 5. 15

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gigawatt

- Believed to relieve Ethiopia's acute energy shortage and to export electricity to neighbouring countries (Kenya, Sudan, and Djibouti) and have deal with others.
- Intended to contribute regional economic integrity.
- The lake being created is the biggest in Ethiopia, have potential for fishing and irrigation development.

Illustrative Examples

13. Climate change can affect agriculture in terms of:

A) Declining soil fertility

B) Increasing water requirement by plants

C) Decreasing crop productivity

D) All are answer

Explanation: Climate change can affect agriculture in terms of: Decline soil fertility, decreasing crop productivity, increasing water requirement by plants, increasing desertification, decrease revenue, etc.

Answer: D

14. Agriculture biodiversity refers to all components of biological diversity, except_____ ?

A) Plant genetic resources

B) Bacteria and fungi genetic resources

C) Animals genetic resources

D) Virus genetic resource

Explanation: Agriculture biodiversity refers to all components of biological diversity relevance to food and agriculture. These are: plants animal, bacteria and fungi.

Answer: D

15. Mitigation of climate hazard involves, except _____ ?

A) Keeping homes away from floodplains,

B) Engineering road and brigades to withstand earthquake,

C) Developing effective building standards

D) Praying on groups during accident

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Explanation: mitigation involves keeping homes away from floodplains, engineering road and bridges withstand earthquake, buying flood insurance, developing effective building standards.

Answer: D

16. Plants genetic resources consists of all, except _____ ?

A) Crops
B) Medicinal plants
C) Pastures
D) Domestic animals

Explanation: Plant genetic resources consist of: wild plants trees no farms, pastures, range land species, medicinal plants, and ornamental plant.

Answer: D

17. Which of the following diseases can be an effect of climate change on human heath?

A) Cholera
B) Asthma
C) Malaria
D) All are answer

Explanation: Climate change cause an increase in the number of heat or cold - related deaths (it could be asthma or respiratory disease), and water - borne illnesses (cholera outbreak) and vectors born (can causes malaria).

Answer: D

18. The immediate reason of sea level rise and flooding in coast and low lying areas is:

A) Greenhouse gases
B) Solar radiation
C) Melting of ice
D) Heavy rain

Explanation: Climate change causes due to rise of greenhouse gases, solar radiation are the main cause to warm the globe, then melting of ice in the polar area, which leads to the rise of sea level. The immediate reason is melting of ice.

Answer: C

19. Which of the following is not true about. The Great Ethiopia Renaissance Dam (GERD)?

A) It is the best example in Ethiopia to support climate resilience green economy

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B) It is a hydroelectric dam on the Abay (Blue Nile) River.

C) It is largest hydroelectric power in Africa

D) Built to anger Egypt ans Sudan

Explanation GERD is Intended to contribute regional economic intercity, NOT to anger Egypt or Sudan.

Answer: D

20. In which international conventions that the first time it was decided that country would propose a natural contribution known as Intended Nationally Determined Contributions (INDC) to reduce emission?

A) Kyoto convention

B) Copenhagen (COP 15)

C) Paris agreement

D) The United Nations Framework Convention on Climate Change

Explanation: For the first time it was decided that each country would propose a national contribution known as intended Nationally Determined Contributions (INDC) to reduce emission. All the 195 UNFCC countries promised to reduce their greenhouse gas emissions by 2025 - 2030.

Answer: B

21. Explain the reason that climate changes the structure of a forest?

Explanation: because the new species that colonize the area create a new type of forest. Thus, once the structure of the forest is changed, it will never be like the existing one.

22. List some of the effects of climate change that causes natural disasters?

Explanation: Hurricanes, Drought, Wildfires, High winds, Flooding etc.

23. List some general preparedness or safety tips that keep you and family safe during a natural disaster to prevent injuries and make the difference in an emergency?

Explanation: Some of the precautions and preparedness during natural disaster includes:

- Stay informed
- Have a plan for evacuation
- Keep emergency kits on hand

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- Avoid unnecessary risk
- Go to the safest area in your home

Unit Review Question

IV. Multiple choice item

1. Which of the following gases contributes to both climate change and thinning of the ozone layer?
   A) CO_2 B) CH_4 C) CFCs D) N_2O
2. Climate change occurs because
   A) Carbon dioxide and other greenhouse gases react chemically to produce excess heat
   B) Earth has too many greenhouse and other glassed buildings
   C) Volcanic eruptions produce large quantities of sulfur and other greenhouse gases
   D) Carbon dioxide and other greenhouse gases trap infrared radiation in the atmosphere
3. What is the term for the natural ability of Earth's atmosphere to trap energy from the sun?
   A) Global warming C) Greenhouse effect
   B) Ozone depletion D) Biological magnification
4. Carbon dioxide is called greenhouses gas because
   A) It is used in greenhouse to increase plant growth
   B) It is transparent to sunlight, but traps heat
   C) It is transparent to heat, but traps sunlight
   D) It is transparent to both sunlight and heat.
5. Which of the following is not the definition of climate change?
   A) Refers to long - term shifts in temperature
   B) It is the global phenomenon of climate transformation
   C) It is a systemic change in the long - term state of the atmosphere over multiple decades
   D) Refers to a monthly shift in temperature and weather - pattern
6. Which one is the natural cause of climate change?

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A) Burning fossil fuels
B) Cutting down forests
C) Farming livestock
D) Solar radiation

7. Which one is the man - made cause of climate change?

A) Volcanic eruption
B) Wild fire
C) Farming livestock
D) Solar radiation

8. Which greenhouse gases in the atmosphere is responsible for the greatest amount global warming?

A) Methane B) CO_2 C) Water vapor D) CFC

9. Which greenhouse gas in the atmosphere absorbs less heat per molecule than the other?

A) Methane B) CO_2 C) Water vapor D) CF_6

10. How much is CO2 accounts for the energy imbalance in the atmosphere and risk of temperature among other gases?

A) 70% B) 50% C) 25% D) 100%

11. Which of the following is not a greenhouse gas?

A) Water vapor
B) Carbon dioxide (CO_2)
C) Methane
D) Nitrogen gas

12. Which of the following is the most abundant gas in the atmosphere?

A) Oxygen
B) Nitrogen oxide
C) Carbon dioxide
D) Methane

13. One of the following is not the natural way of balancing CO_2 in the atmosphere?

A) Respiration
B) Photosynthesis
C) Decomposition
D) Burning of coal

14. Which greenhouse gas is released from high voltage electrical transmission system?

A) HFC B) CO_2 C) Methane D) SF_6

15. Which of the following is the most abundant greenhouse gas in the atmosphere?

A) CO_2 B) H_2O vapor C) N_2O D) PFC

16. Which greenhouse gases cloud be produced from fertilizer and animal

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waster?

A) CO_2 B) H_2O vapor C) N_2O D) PFC

17. Methane is a greenhouse gas produced by all of the following, except?

A) Landfills waste damp C) Rice farming

B) Digestive tracts of cattle D) Burning of fossil fuel

18. Which one of the indirect consequence of man - made climate change?

A) Rising maximum temperatures

B) An increase in haevy precipitation

C) Rising sea level

D) Increasing poverty

19. How does climate change affect biodiversity?

A) Spread of invasive species C) Migration of some species

B) Habitat change D) All are answer

20. Agricultural biodiversity refers to all components of biological diversity, except _____ ?

A) Plant genetic resources

B) Bacteria and fungi genetic resource

C) Animal genetic resource

D) Virus genetic resource

21. Plant genetic resources consists of all, except _____ ?

A) Crops C) Pastures

B) Medicinal plants D) Domestic animals

22. Which of the following is animal genetic resource?

A) Crops C) Pastures

B) Medicinal plants D) Domestic animals

23. Climate change can affect agriculture in terms of:

A) Declining soil fertility

B) Increasing water requirement by plant

C) Decreasing crop productivity

D) All are answer

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24. Climate change can affect forest productivity due to:

A) Align species introduction
B) Insect infestation
C) Wild fire and storm
D) All are answer

25. The immediate reason of sea level rise and flooding in coastal and low lying areas is:

A) Greenhouse gases
B) Solar radiation
C) Melting of ice
D) Heavy rain

26. Which of the following disease can be an effect of climate change on human health?

A) Cholera B) Asthme C) Malaria D) All are answer

27. Climate hazards characterized by a tropical storms caused by atmosphere disturbances a low - pressure area, it is accompanied by strong winds moving as a speed of sixty two Kmph or more?

A) Cyclone
B) Flood
C) Earthquake
D) Tsunami

28. A climate hazards characterized by a series of enormous ocean wave caused by earthquakes underwater, area, it is accompanied

A) Cyclone
B) Flood
C) Earthquake
D) Tsunami

29. A climate hazard characterized by the shifting of tectonic plates under the earth's crust, which are responsible for mass destruction is:

A) Cyclone
B) Flood
C) Earthquake
D) Tsunami

30. A precaution such as, moving into higher ground will help to escape a natural disaster of _____ type?

A) Cyclone
B) Flood
C) Earthquake
D) All

31. When you look at warning sign such as dark, greenish sky, rotating cloud, and roars, the most probable climate

A) Cyclone
B) Flood
C) Earthquake
D) Tsunami

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32. The most common and dangerous natural hazards, which are formed due to a flow of water on areas of land that are usually dry is _____ ?

A) Cyclone
B) Flood
C) Earthquake
D) Tsunami

33. Mitigation of climate hazard involves all except _____ ?

A) Keeping homes away from floodplains
B) Engineering road and bridges standards
C) Developing effective building standards
D) Praying in groups during accident

34. Which of the following is not true about the pairs agreement on climate change?

A) Totally replaced the Kyoto agreement
B) Both developed and developing countries are included
C) Signed in 2015 in Paris
D) Developing countries were not included

35. Which of the following is true about the Kyoto protocol on climate change?

A) Totally replaced the Paris agreement
B) Both developed and developing countries are included
C) Signed in 2015 in Paris
D) Developing countries were not included

36. Which of the is NOT true about green legacy initiative in Ethiopia ?

A) Launched in 2019 with the vision of building green and climate resiliency Ethiopia
B) Targeted to plant 20 billion seedling within a period of four years
C) Encompasses agroforestry, green and renewal of urban areas and integrated water and soil resources management
D) It is funded by united nation agencies

37. Which of the following is not true about The Great Ethiopia Renaissance Dam (GERD)?

A) It is the bast example in Ethiopia to support climate resilience green economy

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B) It is a hydroelectric dam on the Abay (Blue Nile) River.

C) It is the largest hydroelectric power in Africa

D) Built to anger Egypt and Sudan

V. Short Answer

38. A sustained action that reduces or eliminates long term risk to people and property from natural hazards and their effects is called _____?

39. An international convention on climate change, which is signed in 1997 and enter to force in 2005 only for developed countries to reduce gas emission by 5% is _____?

40. During natural disaster _____ is a comprehensive approach involving the identification of threaten through implementation of the proposed mitigation measures?

41. _____ are natural events or agents of disaster in terms of what they may do to human settlements or to the environment in weather cycles?

42. The name of the huge dam that built on Abay river (Nile) and located at Benishangul Gumuz in Ethiopia is _____?

43. List some examples actions that Ethiopia has been implementing within the international policy frameworks on climate change?

VI. Essay type

44. Name at least three greenhouse gases and explain how greenhouse gases contribute to climate change?

45. What are the possible consequences of global warming?

46. Why the heat trapping gases are termed greenhouse gases?

47. Why CO_2 is responsible for the greatest amount global warming than other greenhouse gases?

48. Though water vapor is more abundant than CO_2 in the atmosphere, its greenhouse effect is lower than CO_2. Justify your answer?

49. Explain the three key provisions that were stated in Copenhagen to strengthen the global response to the threat of climate change (COP 15)?

50. How do invasive species affect biodiversity?

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51. Explain the importance of maintaining biodiversity to human?
52. Described how climate change may affect sea level, precipitation patterns, organisms (including humans), and food production?
53. Explain the meaning of Carbon selling (emission trading) between countries, which is stated under Kyoto protocol?
54. In Ethiopia malaria was predominantly high in low land: however, it is expanding into highland areas. What could be possible reason?

Answer to the unit Review Questions

I. Multiple choice

1. C	9. B	17. D	25. C	33. D
2. D	10. A	18. D	26. D	34. D
3. C	11. D	19. D	27. A	35. D
4. B	12. A	20. D	28. D	36. D
5. D	13. D	21. D	29. C	37. D
6. C	14. D	22. D	30. B	
7. C	15. B	23. D	31. A	
8. B	16. C	24. D	32. B	

II. Short Answer

38. Mitigation
39. Kyoto protocol
40. Disaster risk management
41. Climate hazard
42. GERD
43. Ethiopia Green Legacy Initiative, GERD

III. Essay type

44. **Explanation:** Greenhouse gases - carbon dioxide, methane, surface ozone, nitrous oxide, and chlorofluorocarbons - cause the greenhouse effect, in which the atmosphere retains heat and warms Earth's surface. Increased levels of CO_2 other human - produces greenhouse gases in the atmosphere are causing concerns about an enhanced greenhouse effect, which is additional warming produced by increased levels of gases that absorb infrared radiation.

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45. The possible consequences of climate change could be a rise in sea level, flooding drought, reduction in crop production.

46. Because these gases are act likes glass walls of a greenhouse, which trap heat hence are called greenhouse gases

47. Because CO_2 is less abundant but stays in the atmosphere much longer than the others.

48. Because unlike CO_2, H_2O persist in the atmosphere for only a few days to hold heat.

49. Explanation:

1. Holding the global increases of average temperature bellow 2 ^{0}C - 1.5 ^{0}C above pre - industrial level, which is believed to reduce the risk of climate change.
2. Develop mechanisms or adaptations to the adverse effects if climate change
3. Making finance flows consistent with a pathway toward low greenhouse gas emission

50. Invasive species out compete the native species for food, water and space

51. Explanation

- Greater food security
- Opportunities for economic development
- Foundation for new pharmaceuticals

52. Explanation

- During the 21st century, climate change is causing a rise sea level. Precipitation patterns are changing, resulting in more frequent droughts in some areas and more frequent flooding in other areas.
- Biologists think that climate change will cause some species to go extinct, some to be unaffected, and others to expand their number ranges.
- Data linking climate change and human health problems (particularly in developing countries) are accumulating.
- Problems for agriculture include flooding, increased droughts, and

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declining agricultural productivity in tropical and subtropical areas.

53. Through emission trading, countries that emit less than they are allowed can sell this amount to industrialized countries that produce more than should to compensate climate Chang crises.

54. Climate change changes the spread of disease vectors such as anopheles mosquito into highlands, which transfer plasm odium parasite to cause malaria.

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